

FinGPT-Forecaster Re-implemented with FinRobot

In this demo, we set up an agent to emulate the behavior of model in the fingpt-forecaster project with AutoGen, which takes a company's ticker symbol, recent basic financials and market news as input and predict its stock movements.

For detail of the original project, check out [FinGPT-Forecaster!](#) 🔥 [Demo](#), [Medium Blog](#) & [Model](#) on Huggingface 😊!

This is a default bot, for more configurable demo, see [advanced tutorial](#)

```
In [ ]: import autogen
        from finrobot.utils import get_current_date, register_keys_from_json
        from finrobot.agents.workflow import SingleAssistant
```

After importing all the necessary packages and functions, we instantiate a SingleAssistant workflow "Market_Analyst". We also need the config for OpenAI & Finnhub here.

- for openai configuration, rename OAI_CONFIG_LIST_sample to OAI_CONFIG_LIST and replace the api keys
- for finnhub configuration, rename config_api_keys_sample to config_api_keys and replace the api keys

```
In [ ]: # Read OpenAI API keys from a JSON file
        llm_config = {
            "config_list": autogen.config_list_from_json(
                "../OAI_CONFIG_LIST",
                filter_dict={"model": ["gpt-4"]},
            ),
            "timeout": 120,
            "temperature": 0,
        }

        # Register FINNHUB API keys
        register_keys_from_json("../config_api_keys")
```

Define the assistant, and simply start chatting!

```
In [ ]: company = "APPLE"

        assitant = SingleAssistant(
            "Market_Analyst",
            llm_config,
            # set to "ALWAYS" if you want to chat instead of simply receiving the prediciton
            human_input_mode="NEVER",
        )
        assitant.chat(
            f"Use all the tools provided to retrieve information available for {company} upon {get_current_date()}"
            f"with 2-4 most important factors respectively and keep them concise. Most factors should be related to the company's stock price movement."
            f"Then make a rough prediction (e.g. up/down by 2-3%) of the {company} stock price movement."
        )
```

User_Proxy (to Market_Analyst):

Use all the tools provided to retrieve information available for APPLE upon 2024-08-05. Analyze the positive developments and potential concerns of APPLE with 2-4 most important factors respectively and keep them concise. Most factors should be inferred from company related news. Then make a rough prediction (e.g. up/down by 2-3%) of the APPLE stock price movement for next week. Provide a summary analysis to support your prediction.

Use all the tools provided to retrieve information available for APPLE upon 2024-08-05. Analyze the positive developments and potential concerns of APPLE with 2-4 most important factors respectively and keep them concise. Most factors should be inferred from company related news. Then make a rough prediction (e.g. up/down by 2-3%) of the APPLE stock price movement for next week. Provide a summary analysis to support your prediction.

Market_Analyst (to User_Proxy):

***** Suggested tool call (call_jr4Qm35h9k5Xs1QgdUFp04WQ): get_company_profile *****

Arguments:

```
{
  "symbol": "AAPL"
}
```

>>>>>> EXECUTING FUNCTION get_company_profile...

Finnhub client initialized

User_Proxy (to Market_Analyst):

User_Proxy (to Market_Analyst):

***** Response from calling tool (call_jr4Qm35h9k5Xs1QgdUFp04WQ) *****

[Company Introduction]:

Apple Inc is a leading entity in the Technology sector. Incorporated and publicly traded since 1980-12-12, the company has established its reputation as one of the key players in the market. As of today, Apple Inc has a market capitalization of 3183508.78 in USD, with 15334.08 shares outstanding.

Apple Inc operates primarily in the US, trading under the ticker AAPL on the NASDAQ NMS - GLOBAL MARKET. As a dominant force in the Technology space, the company continues to innovate and drive progress within the industry.

Market_Analyst (to User_Proxy):

***** Suggested tool call (call_jr4Qm35h9k5Xs1QgdUFp04WQ): get_company_news *****

Arguments:

```
{
  "symbol": "AAPL",
  "start_date": "2024-08-01",
  "end_date": "2024-08-05",
  "max_news_num": 10
}
```

>>>>>> EXECUTING FUNCTION get_company_news...

Finnhub client initialized

User_Proxy (to Market_Analyst):

User_Proxy (to Market_Analyst):

***** Response from calling tool (call_jr4Qm35h9k5Xs1QgdUFp04WQ) *****

date

headline

summary

0 20240802001600

NEW YORK MARKET CLOSE: Stocks suffer as economic nerves spark sell-off

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1 20240802004211

Apple: Fiscal Q3 Earnings Snapshot

CUPERTINO, Calif. — CUPERTINO, Calif. — Apple Inc. on Thursday reported fiscal third-quarter net income of \$21.45 billion. On a per-share basis, the Cupertino, California-based company said...

2 20240802052024

Apple's AI Spending Looks Light Compared to Rivals In the last month, investors have questioned the tens of billions that firms like Meta Platforms, Microsoft, and Alphabet are pouring into generative-AI infrastructure, especially before the technology has demonstrated significant returns on that investment. At its developer day in June, Apple trumpeted its own gen-AI features that promise to improve Siri in the next iPhone and offer other AI driven features. On Thursday's earnings call, CEO Tim Cook promised that the Apple Intelligence AI features would enrich people's lives.

3 20240802134600

Apple likely to outperform in upcoming launches, says Morgan Stanley

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4 20240802141111

Anti-steering: Apple Faces The Music

On 4 March 2024, Apple was fined €1.84bn by the European Commission for abusing its dominant position in respect of the distribution of music streaming apps to iPhone and iPad users ; specifically...

5 20240802150200

LONDON MARKET MIDDAY: US recession fears dampen mood ahead of payrolls

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6 20240802150300 Wall Street Set To Tumble As Macro Concerns Return, VIX Spikes And Global Markets Sell Off: Safe-Haven Gold Rises To Record High As Traders Eye July Jobs Data

Looking for stock market analysis and research with proven results? Zacks.com offers in-depth financial research with over 30 years of proven results.

7 20240802160200

Stock index futures fall sharply as job growth was less than expected

Looking for stock market analysis and research with proven results? Zacks.com offers in-depth financial research with over 30 years of proven results.

8 20240803212000

Berkshire Hathaway Slashes Apple Stake by Nearly 50%

Berkshire Hathaway slashed its big stake in Apple by nearly 50% in the second quarter to about 400 million shares from 789 million shares in the first quarter, according to information in the company's 10-Q report released Saturday. Berkshire now holds about 2.6% of Apple and the stake was worth about \$88 billion based on Friday's closing price of \$219.86. The reduction in the Apple stake wasn't a surprise but the size of the cut was unexpected.

9 20240804233440

Apple Investors Urged to Stay Calm After Buffett Slashes Stake

(Bloomberg) — To some, Berkshire Hathaway Inc.'s gutting of its Apple Inc. stake could be interpreted as a lack of conviction in the iPhone maker's growth story. But many on Wall Street are urging investors to look past the news and stay calm. Most Read from Bloomberg Singer Akon's Multibillion-Dollar Futuristic City in Africa Gets Final Notice What a Beautiful Bus Stop Can Do Valencia Follows Barcelona in Crackdown on Short-Term Rentals A Vast Wetland Park Seeks to Slake a Thirsty Megacity A Radical

Market_Analyst (to User_Proxy):

***** Suggested tool call (call_jr4Qm35h9k5Xs1QgdUFp04WQ): get_basic_financials *****

Arguments:

```
{
  "symbol": "AAPL",
  "selected_columns": ["eps", "grossMargin", "netMargin", "peTTM", "totalDebtToEquity", "totalDebtToTotalAsset", "totalDebtToTotalCapital"]
}
```

>>>>>> EXECUTING FUNCTION get_basic_financials...

Finnhub client initialized
User_Proxy (to Market_Analyst):

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***** Response from calling tool (call_jr4Qm35h9k5Xs1QgdUFp04WQ) *****
Error: dictionary changed size during iteration

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}

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Error: dictionary changed size during iteration

Market_Analyst (to User_Proxy):

***** Suggested tool call (call_jr4Qm35h9k5Xs1QgdUFp04WQ): get_stock_data *****
Arguments:
{
 "symbol": "AAPL",
 "start_date": "2024-07-29",
 "end_date": "2024-08-05"
}

>>>>>>> EXECUTING FUNCTION get_stock_data...

User_Proxy (to Market_Analyst):

User_Proxy (to Market_Analyst):

***** Response from calling tool (call_jr4Qm35h9k5Xs1QgdUFp04WQ) *****

	Open	High	Low	Close	Volume	Dividends	S
tock Splits							
Date							
2024-07-29 00:00:00-04:00	216.960007	219.300003	215.750000	218.240005	36311800	0.0	
0.0							
2024-07-30 00:00:00-04:00	219.190002	220.330002	216.119995	218.800003	41643800	0.0	
0.0							
2024-07-31 00:00:00-04:00	221.440002	223.820007	220.630005	222.080002	50036300	0.0	
0.0							
2024-08-01 00:00:00-04:00	224.369995	224.479996	217.020004	218.360001	62501000	0.0	
0.0							
2024-08-02 00:00:00-04:00	219.149994	225.600006	217.710007	219.860001	98480100	0.0	
0.0							

Market_Analyst (to User_Proxy):

Positive Developments:

1. Apple's Q3 earnings report showed a net income of \$21.45 billion, indicating strong financial performance.
2. Morgan Stanley predicts that Apple is likely to outperform in the upcoming launches, suggesting potential growth.
3. Apple's AI features are expected to improve Siri and offer other AI-driven features, indicating innovation and potential for increased user engagement.

Potential Concerns:

1. Berkshire Hathaway slashed its stake in Apple by nearly 50%, which could be interpreted as a lack of conviction in Apple's growth story.
2. Apple was fined €1.84bn by the European Commission for abusing its dominant position in respect of the distribution of music streaming apps to iPhone and iPad users, indicating potential legal and regulatory challenges.

Stock Price Prediction:

Given the recent news and developments, it is likely that Apple's stock price may experience a slight decrease in the next week. The reduction of Berkshire Hathaway's stake could lead to short-term selling pressure. However, the strong Q3 earnings and positive outlook from Morgan Stanley may help to mitigate this decrease.

Summary Analysis:

Apple continues to show strong financial performance and innovation in its products. However, recent events such as the reduction of Berkshire Hathaway's stake and the fine from the European Commission may lead to short-term volatility in the stock price. Investors should monitor these developments closely. Despite these concerns, Apple's strong financials and positive outlook suggest that it remains a solid long-term investment.

Please note that this analysis is based on the information available up to August 5, 2024, and future developments may alter this prediction.

User_Proxy (to Market_Analyst):

Market_Analyst (to User_Proxy):

TERMINATE

Current chat finished. Resetting agents ...